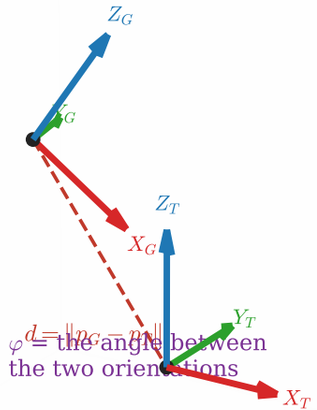
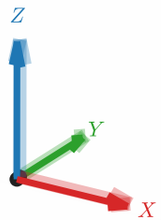
Confondu — the gripper frame and the target frame — the definition every model is scored on

before: two separate frames



confondu: the frames coincide



$d < 0.02$ m and $\varphi < 0.02$ rad (1.15°)

The gripper frame (built in code, not a Transform):

$q_G = R_{\text{fingerLink1}} \cdot \text{Euler}(-90,\ 90,\ 0)$

$p_G = \frac{1}{2}(p_{f_1} + p_{f_2}) + q_G \cdot (0, 0, 0.02)$

(the grasp centre between the two fingers, pushed 2 cm along the gripper's local $+Z$; the Euler offset is the Inspector value)

The target frame:

$q_T = R_{\text{target}} \cdot \text{Euler}(0, 0, 0), \quad p_T = p_{\text{target}}$

(targetFrameOffsetEuler is zero: the curriculum already authors the target with $+Z$ pointing away from the robot)

The two errors the gate is made of:

$d = \|p_G - p_T\|$

$\varphi = 2\arccos\,|\langle q_G, q_T \rangle| \ \in [0, \pi]$

the $|\cdot|$ is the double-cover fix (q and $-q$ are the same rotation); φ is the geodesic angle, so it is frame-independent

What the policy sees (14 numbers, base_link FLU):

[0:3] p_G [3:7] q_G [7:10] p_T [10:14] q_T

quaternions canonicalised $w > 0$ before being observed

Confondu = both errors inside the gate at the same time.

Position alone is not the task: the wrist must arrive aligned.